SCHOOL OF COMPUTER SCIENCE AND IT
YBN UNIVERSITY RANCHI

JAVA PROGRAMMING
ASSIGNMENT 04

1) Reverse a String

Description: Write a Java program that prompts the user to input a string and then prints the string in reverse order.

Hint:

- Use the `charAt()` method to access characters in the string.
- Iterate through the string from the last character to the first and build the reversed string.
- You can also explore using the `StringBuilder` class and its `reverse()` method.

2) Check if a String is a Palindrome

Description: Write a Java program that checks whether a given string is a palindrome (a string that reads the same forward and backward).

Hint:

- Ignore spaces and case sensitivity by converting the string to lower case and removing spaces using `toLowerCase()` and `replaceAll()` methods.
- Compare the string with its reverse to determine if it is a palindrome.

3) Count the Number of Vowels and Consonants in a String

Description: Write a Java program that counts the number of vowels and consonants in a given string.

Hint:

- Use the `charAt()` method to iterate through the string.
- Check each character to see if it is a vowel (a, e, i, o, u) or a consonant by checking if it's a letter and not a vowel.
- Use `Character.isLetter()` to check if a character is a letter.

4) Find the Longest Word in a Sentence

Description: Write a Java program that takes a sentence as input and finds the longest word in that sentence.

Hint:

- Use the `split()` method to break the sentence into words.
- Iterate through the array of words, checking the length of each word to find the longest one.

5) Count the Frequency of Each Character in a String

Description: Write a Java program that counts the frequency of each character in a given string.

Hint:

- Use a `HashMap` to store each character as a key and its frequency as the value.
- Iterate through the string using `charAt()` and update the frequency in the `HashMap`.
- Display the frequency of each character at the end.